

Master Research Plan — v2 (amended)

Parameter Estimation for the Ornstein-Uhlenbeck Process under Irregular Sampling: A Computational Study

Owner: Suryansh Kumar · Proposed guide: Prof. Chetan Pahlajani (IIT Gandhinagar) · Window: Aug 7 - Oct 31, 2026 · ~8 h/week ≈ 96 h

v2 = v1 + due-diligence amendments (Aug 22, 2026) + theory-ladder status + outreach-package state. Companion: w1_w2_execution_doc.md. Sequencing is binding; calendar dates are owner-adjusted.

0 · Standing decisions (locked)

Item	Decision
Research question	Where do standard OU estimators leave their operating envelope — bias and CI-coverage collapse beyond what first-order theory prices — as observation times grow more variable (CV) and coarser ($\theta\Delta$), and can a practitioner locate their dataset relative to that boundary?
Differentiation sentence (v2, post-diligence)	First systematic finite-sample map of 95% CI-coverage collapse for exact / PFML / Euler OU estimators as gap-CV dials 0→3, twin-normalized to isolate the cost of irregularity, delivering an operating-region boundary in the ($\theta\Delta$, CV) plane that the analytic (Åit-Sahalia-Mykland) and bias-focused (astronomy, ecology) literatures do not provide.
Scope	Floor: 1 model (OU) × 3 estimators (E-exact, E-PFML, E-euler) × spacing {equidistant + Gamma CV-dial} × 3 metrics; E1+E2(+E3). Stretch (W8 gate only): MMPP clustering. Future Work: CIR/GBM, real data, microstructure noise, endogenous sampling, diagnostics software, theory lemmas.
Alternative in the email	Monte Carlo variance reduction under fixed computational budgets — Pahlajani's pick overrides everything here.
Governing principles	Finished beats impressive · claims one week behind reality · pre-commit thresholds before headline runs · every figure regenerable by one command.
Finance story (intro)	OU = Vasicek short-rate model; OU = pairs-trading spread; irregular timestamps = transaction-time sampling.

0.5 · Due-diligence outcomes (Aug 22, 2026) — NEW

- **Kill-criterion verdict: does not fire.** No published work combines all three axes (≥ 2 of our estimators compared / spacing distribution varied / empirical CI coverage reported). Two papers hit 2 of 3 axes.

Closest work	What it does	What it does not do
Åit-Sahalia & Mykland 2003, Econometrica	FIML / IOML / PFML (= our E-PFML, verbatim ' Δ in place of actual Δ '); fixed vs random sampling; asymptotic bias/variance expansions + confirmatory sims	No finite-sample CI coverage; no CV dial; no operating boundary
Åit-Sahalia & Mykland 2004, Ann. Statist.	General theory for randomly spaced observations; exact-MLE and Euler contrasts as worked examples	Purely asymptotic; no coverage
Fleming et al. 2017/2019 (ctmm, ecology)	Multiple OU estimators (ML/REML/pHREML) with empirical coverage	Regular spacing only — duration varied, not gap distribution
Holý & Tomanová 2018→2025 (AOR)	OU under UHF microstructure noise; irregular (Poisson) spacing; bias/MSE	Noise axis, not irregularity; no coverage; no CV dial; no naive baseline
Kozłowski 2017 (A&A) + DRW literature	MC over cadence/baseline for OU=DRW; baseline $\geq 10\tau$ rule	Bias of τ only; single estimator; no coverage

- **Four required edits — all incorporated in v2 and in the outreach package:** (1) E-naive renamed **E-PFML** with AS&M-2003 attribution; our coverage study positioned as the finite-sample counterpart of their asymptotic PFML result. (2) **Gap floor** $\Delta_i \leftarrow \max(\Delta_i, \epsilon\Delta)$ for $CV \geq 1.5$, ϵ pre-registered, clip fraction reported per cell. (3) The 94–96% certification band restricted to the certification cell; expected E-exact coverage breach at low-span corners stated up front (order- T^{-1} bias, Tang-Chen 2009; Yu 2012), with twin design attributing it to span, not irregularity. (4) Twin normalization framed as the empirical mirror of AS&M's cost-of-discreteness vs cost-of-randomness decomposition.
- **Optional strengthening (venue ceiling):** a bias-corrected-coverage layer (Tang-Chen/Yu correction or the parametric bootstrap) as a second money figure. **Commitment:** re-run a targeted 2024–2026 preprint scan immediately before submission; if the kill criterion then fires, pivot to MMPP clustering or the bias-corrected angle as primary.

1 · Mathematical framework

1.1 Model

$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$, $\theta > 0$. Exact transition: $X_{t+\Delta} | X_t=x \sim N(\mu + (x-\mu)e^{-\theta\Delta}, (\sigma^2/2\theta)(1 - e^{-2\theta\Delta}))$; stationary law $N(\mu, \sigma^2/2\theta)$; lag- Δ autocorrelation $e^{-\theta\Delta}$. Under gaps Δ_i : a Gaussian AR(1) with $\phi_i = e^{-\theta\Delta_i}$. (Derivations D1-D5: companion doc §1.2; ladder P3-P5/M1.)

1.2 Estimators — all map $(x, \text{times}) \rightarrow (\theta, \hat{\mu}, \hat{\sigma})$

ID	Definition	Expected pathology / role
E-exact	Exact MLE: sum of true Gaussian transition log-densities, gap-dependent coefficients	Benchmark; point estimates clean by design; finite-sample coverage still to be measured
E-PFML	Exact likelihood with every $\Delta_i \rightarrow \Delta$ (= AS&M 2003 PFML; 'ignore the timestamps')	Bias in θ, σ growing with gap variability; our study = finite-sample coverage counterpart of AS&M's asymptotic bias analysis
E-euler	Euler/Gaussian contrast (Florens-Zmirou 1989; AS&M-2004 example): mean $x + \theta(\mu-x)\Delta_i$, variance $\sigma^2\Delta_i$	Discretization bias growing with $\theta\Delta$; interacts with variability

Working parameterization ($\log \theta, \mu, \log \sigma$); L-BFGS-B, 3 multi-starts ($\times 5$ in flagged cells); *expm1* for variance terms.

1.3 Spacing families (matched mean gap Δ)

Family	Construction	Dial / amendment
Equidistant	$\Delta_i = \Delta$	CV = 0 (also the twin baseline)
Gamma (incl. exponential)	$\Delta_i \sim \Gamma(k, s)$, $k = CV^{-2}$, $s = \Delta \cdot CV^2$	$CV \in \{0.25, 0.5, 1 = \text{exponential}, 1.5, 2, 3\}$; gap floor $\Delta_i \leftarrow \max(\Delta_i, \epsilon\Delta)$ for $CV \geq 1.5$, ϵ pre-registered, clip fraction reported
Clustered MMPP — stretch	2-state MMPP; sojourn means ($10\Delta, 2\Delta$); rate calibration to $1/\Delta$	$r = \lambda_b/\lambda_c \in \{1, 5, 20\}$; W8 gate only

1.4 Metrics, uncertainty, normalization

- Per cell: relative bias and relative RMSE for θ, μ, σ ; empirical coverage of 95% Wald CIs — observed Fisher information (numerical Hessian at the optimum in working parameters), inverted, delta method to (θ, μ, σ) with Jacobian $\text{diag}(\theta, 1, \hat{\sigma})$. Parametric bootstrap (500 draws) spot-check on 2 cells.
- Twin normalization (referee armor)**: every irregular cell paired with an equidistant twin (same n , same Δ); report excess bias, RMSE ratio, coverage gap vs the twin — the empirical mirror of AS&M's theoretical decomposition; isolates the cost of *irregularity* by construction. Compare measured inflation against AS&M first-order predictions in the Discussion; never lean on asymptotics as ground truth.

1.5 Pre-registration and certification

- Frozen before any headline run: outside the operating region iff coverage $< 80\%$ (nominal 95%) or $|\text{rel. bias}| > 10\%$ for θ or σ . Money figure F7 = boundary of that region in the $(\theta\Delta, CV)$ plane, one curve per estimator.
- Certification (cell-restricted)**: E-exact on equidistant data must hit 94-96% coverage at $\theta\Delta = 0.5$, $n = 1000$ (band = ± 2 MC-SE at 2000 reps; $SE \approx 0.49\%$). At low-span corners (e.g. $\theta\Delta = 0.1$, $n = 250$, $T = 25$) even E-exact is *expected* to breach 80% via the order- T^{-1} mean-reversion bias (Tang-Chen 2009; Yu 2012) — an expected finding the twin attributes to span; certification failure at the certification cell itself would instead indicate a pipeline bug (Hessian conditioning / Jacobian) and blocks all headline runs.

2 · Theory ladder and current status (NEW section)

Rung	Content	Status (Aug 22)
P1, P1b	Brownian motion: three demands + construction (pre-scaled calibration $\sigma_s^2 = c\delta$); $\text{Cov}(B(s), B(t)) = \min(s, t)$	✓ done
S1, S2	Chain rule of densities; Markov property (PMF + density form); master factorization $p(x_0 \theta)\prod p(x_i x_{i-1}, \theta)$	✓ done
S3	Gaussian AR(1): k-step solution, stationary $N(\mu, \sigma_\epsilon^2/(1-\phi^2))$, $\rho(k)=\phi^k$, conditional MLE	✓ done — MLE part flagged: revisit before M2
P2	SDE notation $\leftrightarrow$ integral equation; worked read-off solutions; OU self-reference identified	✓ done — gate exercise (a)(b)(c) pending; blocks P3
P3	Wiener integral $\int f dB$ (deterministic f) from step functions; Itô isometry; consumes imports L^1, L^3	○ NEXT — solo, ~55 min
P4 + P5	OU solution (integrating factor; $\mu \neq 0$ via $Y=X-\mu$; consumes L^2) + transition density, stationary law, autocorr $e^{-\theta\Delta}$	○ combined, ~65 min; exit test: derive transition density start-to-finish
M1 + M2	Merge (OU at gaps = AR(1), $\phi_i=e^{-\theta\Delta_i}$, $v_i=(\sigma^2/2\theta)(1-e^{-2\theta\Delta_i})$); exact + PFML log-likelihoods	○ ~45 min; includes S3-MLE revisit
E-euler rung	Euler contrast as small- $\theta\Delta$ truncation of the exact transition	○ short, after P5 (promoted: floor estimator)

Rung	Content	Status (Aug 22)
M3 (+ext)	Poisson/exponential gaps (CV=1) + Gamma family $k=CV^{-2}$, $s=\Delta CV^2$	○ ~30 min, anytime before gap generator
S4 (new)	Fisher information, Wald CIs, delta method, parametric bootstrap	○ ~45-60 min, before certification work
C1-C3	Exact simulation; MLE hygiene; Monte Carlo recovery (Week-2 notebook cells)	○ after M2

- **Import ledger (only citable non-derived items):** substrate — BM exists with its stated properties (Wiener measure, out of scope); L^1 — step-sum Gaussians converge (limit Gaussian, parameters pass, disjoint-interval independence survives); L^2 — integration-order swap for deterministic integrands; L^3 — Gaussian closure. **Exit condition:** reproduce D1-D5 on blank paper ≤ 30 min citing only L^1 - L^3 + substrate.
- **Errata caught en route (reported to MIT OCW):** L8 AR(1) stationary variance $\sigma^2/(1-\phi) \rightarrow \sigma^2/(1-\phi^2)$; L17 random-walk construction missing $\sqrt{n}$ normalization (formal disproof written; fix = the calibration). **Reference artifact:** stochastic_processes_for_OU_notes.html (12 sections + problem sets + formula card, current through P2).

3 · Experimental design

Experiment	Grid	Output
E1 — variability dose-response	$\theta\Delta = 0.5$, $n = 1000$; CV 0→3; 3 estimators $\times$ 3 metrics; ≥ 2000 reps/cell	Curves: metric vs CV per estimator (F2-F4), twin-normalized panels
E2 — coarseness $\times$ variability	$\theta\Delta \in \{0.1, 0.5, 1, 2\} \times CV \in \{0, 0.5, 1, 2, 3\} \times n \in \{250, 1000, 4000\}$; ≥ 2000 reps/cell	Heatmaps per estimator/metric (F5-F6); operating-region contours (F7)
E3 — anomaly hunt	Suspect cells at 10k reps; optimizer failures; boundary cells	1-page memo; failure log (failure rates per cell are findings)
E4 — clustering (stretch)	E1 under MMPP, $r \in \{1, 5, 20\}$, CV-matched	F8: coverage vs clustering dial

- Simulation discipline: exact-transition recursion only (never Euler for data generation); default_rng seed spawning (parent per experiment, child per replication); every cell config serialized; one-row-per-replication storage (parquet/CSV) so all figures regenerate from raw.
- True parameters $\theta=1$, $\mu=0$, $\sigma=0.5$ throughout; report invariance to μ, σ rescaling and θ -only-through- $\theta\Delta$, verified on 2 cells.

4 · Week-by-week plan (amended; dates owner-adjusted)

Week	Focus	Key content (amendments in bold)	Exit
W1	Prior art + math	Kill-criterion diligence (done, verdict clear); derivations D1-D5	Diligence log ✓; blank-paper D1-D5 ≤ 30 min
W2	Notebook + email	4-cell notebook, repo, README; outreach package (email + 1-page floor proposal) drafted with all four diligence edits	Fresh clone < 5 min; email sent
W3	Harness v1	spacing.py (gap floor + clip logging), estimators.py (E-exact, E-PFML, E-euler), metrics.py, runner.py, plots.py; smoke 3 $\times$ 3; read AS&M 2003/2004 + Tang-Chen at working depth	One YAML $\rightarrow$ end-to-end run $\rightarrow$ figure
W4	Validation + prereg	S4 theory session (Fisher/Wald/delta/bootstrap); certification at the certification cell only ; freeze preregistration.md (grid, thresholds, reps, seeds, ε , no-peeking)	Certification passes; prereg committed
W5	E1	Run + F2-F4 first pass	Twin-normalized curves plotted
W6	E2	Full grid (largest compute; overnight/VM); F5-F6; failed-cell log	Grid complete; failure log started
W7	E3	Anomaly hunt; 1-page memo; check-in	Memo sent; anomalies resolved/documented
W8	Gate	Floor secure? Yes $\rightarrow$ E4 (MMPP). No $\rightarrow$ buffer. Record verdict	Gate verdict in repo
W9	Figure freeze	Finals F1-F7 (+F8; optional F7b bias-corrected boundary); styling; threshold table T1; no new experiments after	figures/final/ frozen
W10	Draft I	§§1-3; finance installation in intro	Sections drafted
W11	Draft II	§§4-7; claims ledger; full draft to Pahlajani	Draft out for review

Week	Focus	Key content (amendments in bold)	Exit
W12	Ship	Revisions; abstract last; reproducibility pass; pre-submission preprint re-scan ; arXiv (stat.CO; cross-list stat.ME, q-fin.ST)	Preprint live; repo public

Status snapshot (Aug 22): done — diligence + verdict; theory through P2; comprehensive notes artifact; outreach package drafted (send pending: connection line, signature, owner's call). Pending — P2 gate exercise (blocks P3); D1–D5 to blank-paper standard; PS2 submissions A1/A2/A4/B1–B5; notebook.

5 • The paper (~4,500 words + 7-9 figures)

§	Section	Budget	Must contain
—	Abstract (last)	150–200	Armored positioning: degradation beyond first-order theory; one quantitative threshold statement
1	Introduction	700–900	Finance installation (Vasicek 1977; Avellaneda–Lee 2010; transaction-time sampling); other communities one sentence each; 3-bullet contribution list
2	Model & estimators	600–800	D1–D5 condensed; three estimators with pathology intuition; E-PFML attribution
3	Experimental design	500–700	Pre-registration, grids, twin normalization (AS&M-mirror framing), seed policy; cite repo
4	Results	900–1200	E1 curves → E2 maps → anomalies; every claim tied to a figure
5	Practical guidance	400–600	F7 (+F7b); threshold table T1 with validity domain (OU, these grids, Wald CIs)
6	Related work	400–500	AS&M 2003/2004; Tang–Chen; Yu; Holý–Tomanová; Kessler; Florens-Zmirou; Duffie–Glynn; Sørensen; ctmm; DRW; differentiation sentence verbatim
7	Limitations & future	200–300	Everything demoted in §0, one line each

Figures: F1 teaser (PFML vs exact sampling distributions) · F2 coverage vs CV · F3 rel-bias vs CV (θ , σ) · F4 RMSE ratio to twin · F5–F6 coverage heatmaps · F7 operating-region boundaries (money) · optional F7b bias-corrected boundary · F8 stretch clustering · T1 thresholds with validity domain. Writing standards: claims ledger; self-contained captions; no number in text unless regenerated by the final pipeline; if results are boring, push the grid until failure appears — and genuinely extended robustness is itself the reported guidance.

6 • Collaboration protocol (Pahlajani)

- **Outreach package (drafted):** ~250-word email — alumnus introduction (B.Tech 2022), candid purpose (MFE applications; recommendation letter *only if the delivered work earns it in his judgment*), honest status (studied + built framing; no code results yet), the MC-variance-reduction alternative, 30-minute ask — plus the 1-page floor proposal (phased: E2 runs on his judgment) and the diligence log on request.
- Kickoff agenda: scope (present §0–§3, invite cuts/redirects) · the alternative — his pick wins · cadence: fortnightly 30 min + 1-page memo (Done / Found / Stuck / Next / Decisions needed) · authorship: ask openly, co-authorship welcome · venue instincts + arXiv endorsement if needed.
- Redirect handling: accept immediately; spacing.py / estimators.py / runner transfer; re-plan W5–W8 with a 2-week ceiling. No-reply path: follow-up +1 week; silent +2 weeks → proceed solo on the floor (work never blocks on him; only the letter does) and try one warm IITGN channel.

7 • Risk register (amended)

Risk	Trigger	Response
Identical prior study	Diligence (done) or pre-submission re-scan	Cleared Aug 22. If the re-scan fires: pivot to MMPP clustering or bias-corrected-coverage as primary; W2-onward machinery unchanged
Optimizer instability	Non-convergence in extreme cells	Multi-start $\times 5$, log-parameterization, profile μ ; document failure <i>rates</i> per cell — fragility is a finding
Coverage artifacts	Wald CIs poor even equidistant at small n/T	Expected at low-T corners (Tang–Chen/Yu); twin attributes correctly; bootstrap spot-check; certification-cell failure = pipeline bug, blocks runs
High-CV numerical blowups	Near-zero gaps at $CV \geq 1.5$	Gap floor $\epsilon\Delta$ (pre-registered) + clip-fraction reporting; sensitivity check on ϵ in E3
Behind at W8 gate	Floor incomplete	Cut in order: E4 → grid resolution → n -sweep to $n=1000$ only. Never cut: certification, thresholds, coverage metric
Guide redirects topic	Kickoff	§6 redirect handling
Compute crunch (W6)	Grid too slow	Vectorize likelihood across replications; overnight; one cloud-VM day; reps to 1000 with a note, never below
Scope creep	Any 'wouldn't it be nice'	Future Work paragraph, not the pipeline; re-read §0

8 • Definition of done

Three estimators $\times \geq 2$ spacing families $\times 3$ metrics, twin-normalized · certification passed at the certification cell and pre-registration honored · ≥ 1 threshold statement with validity domain · E-PFML attribution present · figures F1-F7 (+T1, +F7b/F8 if run) frozen and regenerable by one command · pre-submission preprint re-scan clean · full draft reviewed by Pahlajani · preprint on arXiv (stat.CO primary; cross-list stat.ME, q-fin.ST — confirm with him) · repo public with pinned environment.

9 • Immediate next actions

- **1.** P2 gate exercise (a)(b)(c) — the only item blocking P3.
- **2.** Theory sprint: P3 solo $\rightarrow$ P4+P5 combined (exit test: transition density start-to-finish) $\rightarrow$ M1+M2 (with S3-MLE revisit) $\rightarrow$ notebook Cell 1 same day.
- **3.** PS2 submissions A1, A2, A4, B1-B5 (A3 solved in-session).
- **4.** Outreach: fill connection line + signature in the email; send at owner's chosen moment; S4 session before certification work.

Deliverables inventory: stochastic_processes_for_OU_notes.html · OU_floor_proposal_Suryansh_Kumar.pdf/.docx · email_to_pahlajani.txt · due-diligence report (research artifact) · this plan (OU_master_plan_v2.pdf).